The list Ramsey number of the family of graphs with chromatic number greater than s equals $s^k + 1$

github.com/graph-theory-AI
Machine-generated note (see provenance box)

9 September 2026

Provenance

The mathematics in this note was produced without human intervention, in three stages. (1) The proof was found by the language model `gpt-5.6-sol` in a single autonomous pass on 2026-08-31, as part of a large sweep over open problems catalogued from arXiv (catalog id `2103.15175__00`; original writeup in `attacks/2103.15175__00/output.md`). (2) An independent adversarial referee agent (`claude-fable-5`) re-derived every step, checked the definitions against the source paper and against the paper of Alon, Bucić, Kalvari, Kuperwasser and Szabó, and verified the theorem exhaustively for $s = k = 2$ (all 97,191 list assignments on K_4 up to colour isomorphism, and all 1024 colourings of K_5 with constant lists) and the greedy construction on adversarial and random instances up to 27 vertices; its verdict was CONFIRMED. Its report is reproduced verbatim in Appendix B. (3) On 2026-09-09 the writeup was rewritten into the present note by `claude-fable-5.1`: the text was reorganised with full proofs in Appendix A, definitions were spelled out, and context on the source paper and on the referee's remarks was added as Section 3. No mathematical content was changed; the two degenerate-case observations in Remark 3.3 are taken from the referee's report. **No human mathematician has checked this argument.** We circulate it to obtain exactly that: is the proof correct, and is the result new?

Abstract

The list Ramsey number $R_\ell(\mathcal{H}, k)$ of a family $\mathcal{H}$ of graphs is the least n for which some assignment of k-element colour lists to the edges of K_n forces every colouring from the lists to contain a monochromatic member of $\mathcal{H}$. Fox, He, Luo and Xu proved $\frac{1}{e}s^k \le R_\ell(\mathcal{H}_s, k) \le s^k + 1$ for the family $\mathcal{H}_s$ of graphs with chromatic number greater than s, and conjectured that the upper bound is the truth. We prove the conjecture: $R_\ell(\mathcal{H}_s, k) = s^k + 1$ for all $s \ge 2$ and $k \ge 1$. The new ingredient is a one-paragraph greedy argument: on at most s^k vertices, one can assign to every vertex and every colour a label in $[s]$ so that each edge is separated, in one of its listed colours, by the labels of its ends; colouring each edge with such a colour makes every colour class properly s-coloured.

1 Introduction

For a positive integer m write $[m] = \{1, \dots, m\}$. A *k-list assignment* on a graph K_n is a map L from $E(K_n)$ to the k-element subsets of a set of colours (in the literature, $L : E(K_n) \to \binom{\mathbb{N}}{k}$). An *$L$-colouring* is a map $\varphi : E(K_n) \to \mathbb{N}$ with $\varphi(e) \in L(e)$ for every edge e; its *colour classes* are the spanning subgraphs G_c formed by the edges of colour c. A *monochromatic copy* of a graph H is a subgraph of some G_c isomorphic to H.

List Ramsey numbers were introduced by Alon, Bucić, Kalvari, Kuperwasser and Szabó [1]. For a family $\mathcal{H}$ of graphs, following Fox, He, Luo and Xu [2, Section 3.1],

$$R_\ell(\mathcal{H}, k) = \min\{\, n : \text{there is a } k\text{-list assignment } L \text{ on } K_n \text{ such that every } L\text{-colouring of } K_n \text{ contains a monochromatic copy of some } H \in \mathcal{H} \,\}.$$

Such an L is called *forcing*. Taking all lists equal to $[k]$ shows $R_\ell(\mathcal{H}, k) \leq R(\mathcal{H}, k)$, the classical k-colour Ramsey number of the family.

For $s \geq 2$ let $\mathcal{H}_s$ be the family of all finite graphs with chromatic number greater than s. The classical value $R(\mathcal{H}_s, k) = s^k + 1$ goes back to Erdős, McEliece and Taylor (see [2]). Fox, He, Luo and Xu prove [2, Theorem 8]

$$\tfrac{1}{e}\, s^k \;\leq\; R_\ell(\mathcal{H}_s, k) \;\leq\; s^k + 1$$

and write: “We conjecture that the upper bound in Theorem 8 is the truth.” In the words of the problem catalogue:

Conjecture 1.1 ([2, Section 3.1])**.** For $s \geq 2$, if $\mathcal{H}_s$ is the family of graphs with chromatic number greater than s, then $R_\ell(\mathcal{H}_s, k) = s^k + 1$.

We prove Conjecture 1.1 for all $s \geq 2$ and $k \geq 1$.

Theorem 1.2 (Main theorem)**.** *Let $s \geq 2$ and $k \geq 1$. Then $R_\ell(\mathcal{H}_s, k) = s^k + 1$. More precisely:*

(i) for every $n \leq s^k$ and every k-list assignment L on K_n there is an L-colouring of K_n all of whose colour classes have chromatic number at most s, so $R_\ell(\mathcal{H}_s, k) > s^k$;

(ii) the constant list assignment $L(e) = [k]$ on K_{s^k+1} is forcing, so $R_\ell(\mathcal{H}_s, k) \leq s^k + 1$.

Idea of proof. Part (ii) is the classical product-colouring argument: if all k colour classes were properly s-colourable, the k-tuple of colours of a vertex would be an injection of $s^k + 1$ vertices into $[s]^k$. Part (i) rests on the *separation lemma* (Lemma 2.1): for $n \leq s^k$ vertices and any k-lists, there are labellings $f_c : V \to [s]$, one per colour c, such that every pair uv is separated ($f_c(u) \neq f_c(v)$) in at least one colour $c \in L(uv)$. Colouring each edge with such a colour makes f_c a proper s-colouring of the class G_c, so no class contains a graph of chromatic number greater than s. The lemma is proved greedily, one vertex at a time: when the j-th vertex receives its whole vector of labels $(f_c(v_j))_c \in [s]^{\mathcal{C}}$, each earlier vertex v_i rules out exactly $s^{|\mathcal{C}|-k}$ vectors (those agreeing with v_i on the k coordinates of $L(v_iv_j)$), and $(j-1)s^{|\mathcal{C}|-k} \leq (s^k - 1)s^{|\mathcal{C}|-k} < s^{|\mathcal{C}|}$, so a good vector remains. *[full proof]*

2 The separation lemma

Lemma 2.1 (Separation lemma)**.** *Let $s \geq 2$, $k \geq 1$, let V be a set of $n \leq s^k$ vertices, and let each pair $uv \in \binom{V}{2}$ be assigned a k-element set $L(uv)$ of colours. Put $\mathcal{C} = \bigcup_{uv} L(uv)$. Then there are maps $f_c : V \to [s]$, $c \in \mathcal{C}$, such that for all distinct $u, v \in V$ some $c \in L(uv)$ satisfies $f_c(u) \neq f_c(v)$.*

Idea of proof. Order $V = \{v_1, \dots, v_n\}$ and choose the vectors $x_j = (f_c(v_j))_{c \in \mathcal{C}} \in [s]^{\mathcal{C}}$ successively. At step j the “bad” vectors for an earlier vertex v_i are those agreeing with x_i on the k coordinates in $L(v_iv_j)$; there are $s^{|\mathcal{C}|-k}$ of them. The union of the $j-1$ bad sets has at most $(s^k - 1)s^{|\mathcal{C}|-k} < s^{|\mathcal{C}|}$ elements, so some x_j avoids them all. Vectors are never changed later, so separations persist. *[full proof]*

3 Remarks

Remark 3.1 (Why the lossless bound). Fox, He, Luo and Xu obtain their lower bound $\frac{1}{e}s^k$ from a random-homomorphism argument in which independent maps $\phi_c : V \to V(G)$ are chosen for all colours at once and all $\binom{n}{2}$ edge events must succeed simultaneously; the Lovász Local Lemma then costs the factor e. The argument above instead exposes the vertices one at a time and chooses each vertex's entire colour vector at once, so the union bound runs only over the $j-1$ earlier vertices, and for the target K_s the failure event in a given colour is exactly label equality, of probability $1/s$. This is the referee's explanation (Appendix B) of why the sequential bound is lossless for the chromatic family.

Remark 3.2 (Interpretation and list convention). The proof uses that each list consists of k *distinct* colours, i.e. $L(e) \in \binom{\mathbb{N}}{k}$, which is the definition in both [1] and [2]; the referee confirmed both definitions verbatim. Under a multiset convention the count $|B_i| = s^{q-k}$ in the proof of Lemma 2.1 would change. The quantifier structure of R_ℓ (least n admitting *some* forcing assignment) is exactly what part (i) of Theorem 1.2 refutes for $n = s^k$: no assignment on K_{s^k} is forcing.

Remark 3.3 (Degenerate cases). For $n \leq 1$ there are no pairs and Lemma 2.1 is vacuous; whenever $n \geq 2$ there is at least one edge, so $q = |\mathcal{C}| \geq k$ and the exponent $q - k$ is non-negative. Isolated vertices in a colour class do not affect its chromatic number. In part (ii), the class G_i with $\chi(G_i) > s$ is itself a member of $\mathcal{H}_s$, since $\mathcal{H}_s$ consists of *all* graphs of chromatic number greater than s. (These observations are the referee's.)

Remark 3.4 (Computational checks). The referee's scripts (`verification/scripts/2103.15175__00/`) verify, independently of the greedy argument, that every one of the 97,191 colour-isomorphism classes of 2-list assignments on $E(K_4)$ admits a colouring from the lists with both colour classes bipartite, and that none of the 1024 colourings of $E(K_5)$ from the constant list $\{1, 2\}$ has both classes bipartite; hence $R_\ell(\mathcal{H}_2, 2) = 5 = 2^2 + 1$. The greedy of Lemma 2.1 was also run for $(s, k) \in \{(2, 2), (2, 3), (3, 2), (2, 4), (3, 3), (5, 2)\}$ on s^k vertices, on the adversarial constant-list assignment and on 3200 random assignments, and never got stuck.

Remark 3.5 (Novelty). The argument is one page of elementary counting, which is striking given that the authors of [2] posed the conjecture together with a Local-Lemma proof of the bound weaker by a factor e. The referee scrutinised the proof specifically for this reason and found no flaw, but notes that the simplicity raises the prior that the argument is folklore or exists unpublished. Its searches (arXiv full text, Semantic Scholar citations of [2], web searches for a resolution, September 2026) found no written source; only two papers cite [2], neither on this question. This has not been checked by a human.

A Full proofs

Lemma 2.1 (Separation lemma). *Let $s \geq 2$, $k \geq 1$, let V be a set of $n \leq s^k$ vertices, and let each pair $uv \in \binom{V}{2}$ be assigned a k-element set $L(uv)$ of colours. Put $\mathcal{C} = \bigcup_{uv} L(uv)$. Then there are maps $f_c : V \to [s]$, $c \in \mathcal{C}$, such that for all distinct $u, v \in V$ some $c \in L(uv)$ satisfies $f_c(u) \neq f_c(v)$.*

Proof. Write $V = \{v_1, \dots, v_n\}$ and $q = |\mathcal{C}|$. We choose successively vectors

$$x_j = (f_c(v_j))_{c\in\mathcal{C}} \in [s]^{\mathcal{C}}, \qquad j = 1, \dots, n,$$

where x_1 is arbitrary. Suppose $x_1, \dots, x_{j-1}$ have been chosen, so that $f_c(v_i)$ is defined for all $c \in \mathcal{C}$ and $i < j$. For each $i < j$ let

$$B_i = \{x \in [s]^{\mathcal{C}} : x_c = f_c(v_i) \text{ for every } c \in L(v_i v_j)\}.$$

Thus B_i is exactly the set of choices of x_j that fail to separate v_i from v_j in every colour listed on the pair $v_i v_j$: $x_j \notin B_i$ if and only if some $c \in L(v_i v_j)$ has $f_c(v_j) \neq f_c(v_i)$.

Because $L(v_i v_j)$ consists of k distinct colours, B_i fixes exactly k of the q coordinates and leaves the remaining $q - k$ free, so $|B_i| = s^{q-k}$. Since $j \leq n \leq s^k$,

$$\Big|\bigcup_{i<j} B_i\Big| \;\leq\; (j-1)\, s^{q-k} \;\leq\; (s^k - 1)\, s^{q-k} \;=\; s^q - s^{q-k} \;<\; s^q \;=\; \big|[s]^{\mathcal{C}}\big|.$$

Hence some vector $x_j \in [s]^{\mathcal{C}}$ lies outside every B_i, $i < j$; choose it. Then v_j is separated from each earlier vertex v_i in at least one colour of $L(v_i v_j)$. Vectors already chosen are never modified, so this separation persists to the end. After step n every pair $\{v_i, v_j\}$ with $i < j$ has been separated at step j, and each f_c is a total map $V \to [s]$. □

Theorem 1.2 (Main theorem). *Let $s \geq 2$ and $k \geq 1$. Then $R_\ell(\mathcal{H}_s, k) = s^k + 1$. More precisely:*

(i) for every $n \leq s^k$ and every k-list assignment L on K_n there is an L-colouring of K_n all of whose colour classes have chromatic number at most s, so $R_\ell(\mathcal{H}_s, k) > s^k$;

(ii) the constant list assignment $L(e) = [k]$ on K_{s^k+1} is forcing, so $R_\ell(\mathcal{H}_s, k) \leq s^k + 1$.

Proof. *(i) Lower bound.* Let $n \leq s^k$ and let L be a k-list assignment on K_n, with vertex set V. Apply Lemma 2.1 to obtain maps $f_c : V \to [s]$, $c \in \mathcal{C} = \bigcup_e L(e)$. For each edge uv choose a colour

$$\varphi(uv) \in L(uv) \qquad \text{with} \qquad f_{\varphi(uv)}(u) \neq f_{\varphi(uv)}(v),$$

which exists by the lemma. Then φ is an L-colouring of K_n. For a colour c, every edge uv of the colour class G_c satisfies $f_c(u) \neq f_c(v)$, so f_c is a proper s-colouring of G_c and $\chi(G_c) \leq s$. Every subgraph of G_c has chromatic number at most $\chi(G_c) \leq s$, so G_c contains no copy of a graph with chromatic number greater than s: the colouring φ has no monochromatic member of $\mathcal{H}_s$. As L was arbitrary, no k-list assignment on K_{s^k} (or on fewer vertices) is forcing, and therefore $R_\ell(\mathcal{H}_s, k) > s^k$.

(ii) Upper bound. On K_{s^k+1} give every edge the list $L(e) = [k]$, and let φ be any L-colouring, with colour classes $G_1, \ldots, G_k$. Suppose for contradiction that $\chi(G_i) \leq s$ for every $i \in [k]$, and choose proper colourings $g_i : V(K_{s^k+1}) \to [s]$ of the G_i. Define

$$g(v) = (g_1(v), \ldots, g_k(v)) \in [s]^k.$$

If $u \neq v$, the edge uv has some colour $i = \varphi(uv)$, so $uv \in E(G_i)$ and properness of g_i gives $g_i(u) \neq g_i(v)$; hence $g(u) \neq g(v)$ and g is injective. This is impossible, since $|V(K_{s^k+1})| = s^k+1 > s^k = |[s]^k|$. Therefore some class G_i has $\chi(G_i) > s$, i.e. G_i is a monochromatic member of $\mathcal{H}_s$. Since φ was arbitrary, the constant assignment is forcing and $R_\ell(\mathcal{H}_s, k) \leq s^k + 1$.

Combining (i) and (ii), $R_\ell(\mathcal{H}_s, k) = s^k + 1$. □

B Report of the LLM referee (verbatim)

The following is the unedited report of the adversarial referee agent (`claude-fable-5`, 2026-09-02/03) on the *original* writeup. Its step numbers refer to that writeup, not to this note. The scripts it mentions are in `verification/scripts/2103.15175__00/`. Treat it as machine output, not as an authority.

B.1 Interpretation

The catalog conjecture (Fox, He, Luo, Xu, "Multicolor list Ramsey numbers grow exponentially", arXiv:2103.15175, Section 3.1) is: for $s \geq 2$, if $\mathcal{H}_s$ is the family of graphs with chromatic number greater than s, then $R_\ell(\mathcal{H}_s, k) = s^k + 1$.

I fetched the source paper (arXiv:2103.15175v2) and read Section 3.1 in full. The paper defines, for a family $\mathcal{H}$: "$R_\ell(\mathcal{H}, k)$ is the minimum n such that there exists an $L : E(K_n) \to \binom{\mathbb{N}}{k}$ for which every L-coloring of K_n contains a monochromatic copy of some $H \in \mathcal{H}$." Theorem 8 states $\frac{1}{e}s^k \leq R_\ell(\mathcal{H}_s, k) \leq s^k + 1$ and the paper says verbatim: "We conjecture that the upper bound in Theorem 8 is the truth." I also checked the original definition in Alon–Bucić–Kalvari–Kuperwasser–Szabó ("List Ramsey numbers"): lists are k-element subsets of $\mathbb{N}$ assigned to edges of K_n, an L-coloring picks each edge's color from its list, and R_ℓ asks for the least n admitting a forcing list assignment.

The writeup formalizes exactly this: lower bound = "for every k-list assignment on $E(K_{s^k})$ there is an L-coloring in which every color class is s-colorable" (hence no monochromatic member of $\mathcal{H}_s$, hence no forcing assignment on s^k vertices, hence $R_\ell > s^k$); upper bound = the constant assignment $L(e) = \{1, \ldots, k\}$ forces on K_{s^k+1}. This is the intended reading; there is no interpretation gaming. A reasonable author of the source paper would consider this conjecture resolved by the writeup's statement, if correct — and I found it correct.

B.2 Step-by-step findings

step	label	note
1. Statement / definitional setup	VALID	Matches the FHLX / ABKKS definition of R_ℓ for families; lists are sets of k distinct colors ($\binom{\mathbb{N}}{k}$), exactly as the writeup assumes and flags in its caveat.
2. Separation lemma (statement)	VALID	For $n \le s^k$ and any k-lists on $\binom{V}{2}$, maps $f_c : V \to [s]$ exist separating every pair in some listed color. Well-posed; $\mathcal{C}$ is finite; $q = \lvert\mathcal{C}\rvert \ge k$ whenever $n \ge 2$ (for $n \le 1$ the lemma is vacuous).
3. Greedy construction of $x_j \in [s]^{\mathcal{C}}$; definition of bad sets B_i	VALID	When x_j is chosen, the vectors $x_1, \dots, x_{j-1}$ (hence the values $f_c(v_i)$) are already fixed, so B_i is well defined. $x_j \notin B_i$ is exactly "some $c \in L(v_i v_j)$ has $f_c(v_j) \ne f_c(v_i)$". Once an edge is separated it stays separated: vectors are never modified later.
4. $\lvert B_i\rvert = s^{q-k}$	VALID	B_i fixes the k coordinates indexed by the k distinct colors of $L(v_i v_j)$ and leaves the other $q-k$ free. Re-derived: correct.
5. Union bound $\lvert\bigcup_{i<j} B_i\rvert \le (j-1)s^{q-k} \le (s^k-1)s^{q-k} < s^q$	VALID	$j \le n \le s^k$ gives $j-1 \le s^k-1$, and $(s^k-1)s^{q-k} = s^q - s^{q-k} < s^q$. Strict inequality holds, so a good x_j exists at every step. This is the crux, and it is airtight.
6. Inductive conclusion of the lemma	VALID	Every pair $\{v_i, v_j\}$, $i < j$, is handled at step j and never broken. All f_c are total functions $V \to [s]$.
7. Lower bound: choice of $\varphi(uv) \in L(uv)$ with $f_{\varphi(uv)}(u) \ne f_{\varphi(uv)}(v)$	VALID	Exists by the lemma; φ is a legitimate L-coloring.

step	label	note
8. Each color class G_c satisfies $\chi(G_c) \le s$, so no monochromatic member of $\mathcal{H}_s$; hence $R_\ell(\mathcal{H}_s, k) > s^k$	VALID	f_c is by construction a proper s-coloring of G_c. Any subgraph of G_c has chromatic number $\le \chi(G_c) \le s$, so G_c contains no copy of any graph with $\chi > s$. Since L was arbitrary,no list assignment on K_{s^k} forces, which under the min-n definition gives $R_\ell > s^k$. Quantifier structure correct.
9. Upper bound via constant lists and the product-coloring injection	VALID	On K_{s^k+1} with $L(e) = \{1, \dots, k\}$, if all classes had $\chi(G_i) \le s$, the map $g(v) = (g_1(v), \dots, g_k(v))$ would be injective into $[s]^k$ ($u \ne v$ share an edge of some color i, and $g_i(u) \ne g_i(v)$), contradicting $s^k + 1 > s^k$. So some G_i has $\chi > s$; G_i is itself a monochromatic member of $\mathcal{H}_s$. This is the classical Erdős–McEliece–Taylor argument giving $R(\mathcal{H}_s, k) = s^k + 1$; the writeup reproves it self-contained, correctly.
10. Combination $R_\ell(\mathcal{H}_s, k) = s^k + 1$	VALID	$> s^k$ and $\le s^k + 1$ give equality. Holds for all $s \ge 2$, $k \ge 1$.

No GAP or ERROR found at any step. I attempted every standard failure mode (off-by-one in the union bound, quantifier slips between “exists a forcing L” and “for all L”, multiset vs. set lists, separation constraints broken by later greedy choices, subgraph-closure of the avoidance property) and none applies.

Why the original authors plausibly missed this: their lower bound machinery (Lemma 6 of the paper) chooses independent random homomorphisms $\phi_c : V \to V(G)$ for all colors and needs all $\binom{n}{2}$ edge events to succeed simultaneously, which forces the Lovász Local Lemma and costs the factor e ($c_2 = 1/e$). The writeup instead exposes vertices one at a time, choosing each vertex’s entire color-vector at once, so the union bound is only over the $j-1$ earlier neighbors, not over all edges — and for the specific target $G = K_s$ the per-color failure event is exactly label-equality with probability exactly $1/s$, making the sequential bound lossless. A simple but genuine improvement, special to the chromatic family.

B.3 Reference check

The writeup is self-contained and cites no external theorems; the only external dependency is the definition of R_ℓ. Verified:

- arXiv:2103.15175v2 (Fox–He–Luo–Xu), fetched in full (7 pp.). Section 3.1 definition of $R_\ell(\mathcal{H}, k)$ for families, Theorem 8 ($\frac{1}{e}s^k \le R_\ell(\mathcal{H}_s, k) \le s^k+1$, $s \ge 2$), and the sentence “We conjecture that the upper bound in Theorem 8 is the truth" all confirmed verbatim. The writeup's formalization matches with identical hypotheses.

- Alon–Bucić–Kalvari–Kuperwasser–Szabó, "List Ramsey numbers" (J. Graph Theory 96 (2021)), fetched: definition $R_\ell(G,k) = \min\{n \mid \exists L : E(K_n^{(r)}) \to \binom{\mathbb{N}}{k}$ s.t. every L-coloring has a monochromatic copy$\}$ — matches.
- Prior-art search for ALREADY_KNOWN: an arXiv full-text query for "list Ramsey" returns only 4 papers (the two above, the double-stars paper arXiv:2211.03642, and the list-Ramsey random-graph threshold paper, CPC 2024); Semantic Scholar lists only 2 citing papers for 2103.15175 (double stars; sharp thresholds for Ramsey properties). None resolves this conjecture. Web searches for a resolution of $R_\ell(\mathcal{H}_s, k) = s^k + 1$ (2022–2026) found nothing, consistent with the catalog's own 2026-05-15 auto-review. I could not find this result in the literature.

B.4 Computational check

Scripts in `verification/scripts/2103.15175__00/`:

1. `exhaustive_k4.py` — exhaustive verification of the lower-bound theorem at $s = 2, k = 2$, $n = s^k = 4$: enumerated **all 97,191** 2-list assignments on $E(K_4)$ up to color isomorphism (restricted-growth canonical form) and for each brute-forced all $2^6 = 64$ L-colorings. **Every one of the 97,191 assignments admits an L-coloring with all color classes bipartite** (0 failures). This check is independent of the writeup's greedy method. Cross-checked by direct (non-canonicalized) enumeration over fixed palettes of sizes 2–5: 1 + 729 + 46,656 + 1,000,000 assignments, 0 failures.
2. `upper_bound_k5.py` — all $2^{10} = 1024$ colorings of $E(K_5)$ with the constant list $\{1,2\}$: **0 of 1024** have both classes bipartite, so the constant assignment forces, confirming $R_\ell(\mathcal{H}_2, 2) \leq 5$. Sanity: on K_4, 18 of 64 colorings have both classes bipartite. Together: $R_\ell(\mathcal{H}_2, 2) = 5 = 2^2 + 1$, exactly as claimed.
3. `greedy_lemma.py` — independent implementation of the writeup's greedy for $(s,k) \in \{(2,2), (2,3), (3,2), (2,4), (3,3), (5,2)\}$ on $n = s^k$ vertices (up to $n = 27$), on the adversarial constant-list assignment (the extremal case where only one free vector remains at the last vertex) plus 3,200 random list assignments over palettes from size k up to $k\binom{n}{2}$: the greedy **never got stuck**,the produced maps f_c separated every edge, and each color class was verified edge-by-edge to be properly s-colored. 0 failures.

All claimed properties verified; no computational check failed.

B.5 Caveats

- **Simplicity vs. open status.** The proof is one page of elementary counting; it is striking that the authors who posed the conjecture (with a proof of the e-weaker bound via the Local Lemma) did not find it. I scrutinized the argument specifically for this reason and found no flaw; the vertex-exposure greedy genuinely sidesteps the LLL loss for the chromatic family (see step 6 discussion). This raises the prior that the argument is folklore or exists unpublished, but no written source was found (searches September 2026).
- **List convention.** The proof needs lists of k *distinct* colors ($|L(e)| = k$), which is the standard $\binom{\mathbb{N}}{k}$ definition in both source papers. The writeup itself flags this. Under a multiset convention the count $|B_i| = s^{q-k}$ would change; that convention is not the one in the literature.
- **Parameter range.** Proven for all $s \geq 2$, $k \geq 1$, matching Theorem 8's hypothesis $s \geq 2$. Degenerate cases are harmless: for $n \leq 1$ the lemma is vacuous; $q \geq k$ holds whenever there is at least one edge; isolated vertices in a color class do not affect its chromatic number.
- **Family membership in the upper bound.** The argument uses that G_i itself (with $\chi(G_i) > s$) is a member of $\mathcal{H}_s$; since $\mathcal{H}_s$ is *all* graphs of chromatic number $> s$, this is immediate.

- The verdict block's one-line claim matches exactly what is proven; no gap between claim and proof.

B.6 Referee summary

I set out expecting to find the error and did not find one. The interpretation is exactly the Fox–He–Luo–Xu conjecture (their wording and definitions verified against arXiv:2103.15175v2 and the ABKKS paper). Every step re-derives correctly: the bad sets have size exactly s^{q-k}, the union bound $(s^k-1)s^{q-k} < s^q$ is strict, the greedy invariant is preserved, the resulting coloring's classes are properly s-colored, and the upper bound is the classical product-coloring argument. Exhaustive computation over all 97,191 color-isomorphism classes of list assignments on K_4 (plus direct enumeration over palettes up to size 5) confirms the theorem at $s = 2, k = 2$, brute force on K_5 confirms tightness, and the greedy was machine-verified on adversarial and random instances up to $n = 27$. No prior resolution exists in the indexed literature. The writeup proves the conjecture $R_\ell(\mathcal{H}_s, k) = s^k + 1$.

References

[1] N. Alon, M. Bucić, T. Kalvari, E. Kuperwasser and T. Szabó, List Ramsey numbers, *J. Graph Theory* 96 (2021) 109–128.

[2] J. Fox, X. He, S. Luo and M. W. Xu, Multicolor list Ramsey numbers grow exponentially, arXiv:2103.15175 (2021; v2 January 2022).